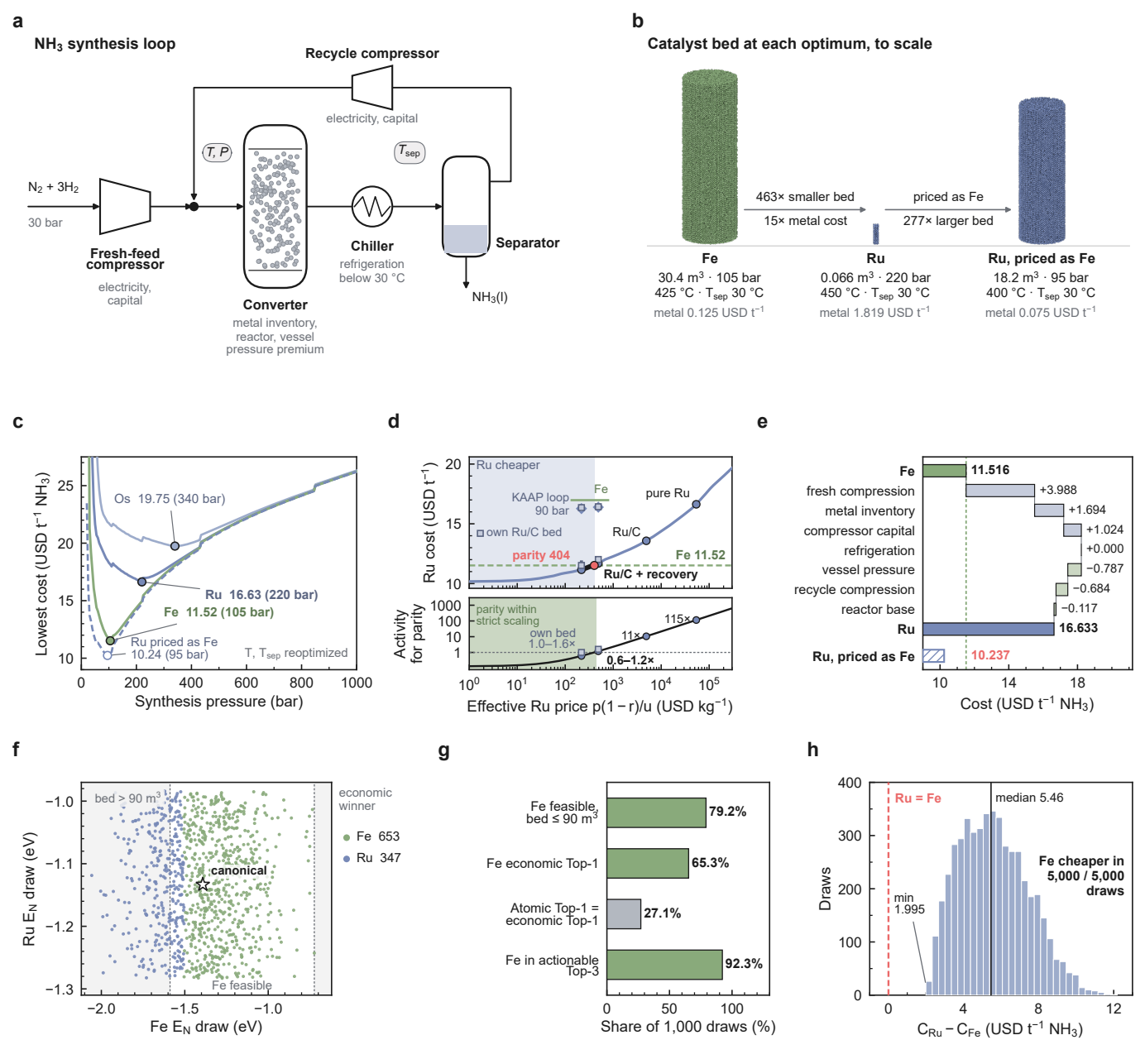

a
NH3 synthesis loop
Recycle compressor
electricity, capital
T, P
Tsep
N2 + 3H2
30 bar
Fresh-feed compressor
electricity, capital
Converter
metal inventory, reactor, vessel pressure premium
Chiller
refrigeration below 30 °C
Separator
NH3(l)
b
Catalyst bed at each optimum, to scale
463× smaller bed
15× metal cost
priced as Fe
277× larger bed
Fe
30.4 m3 · 105 bar
425 °C · Tsep 30 °C
metal 0.125 USD t−1
Ru
0.066 m3 · 220 bar
450 °C · Tsep 30 °C
metal 1.819 USD t−1
Ru, priced as Fe
18.2 m3 · 95 bar
400 °C · Tsep 30 °C
metal 0.075 USD t−1
c
Lowest cost (USD t−1 NH3)
Os 19.75 (340 bar)
Ru 16.63 (220 bar)
Fe 11.52 (105 bar)
Ru priced as Fe
10.24 (95 bar)
T, Tsep reoptimized
Synthesis pressure (bar)
d
Ru cheaper
KAAP loop
90 bar
Fe
pure Ru
own Ru/C bed
Ru/C
parity 404
Fe 11.52
Ru/C + recovery
Ru cost (USD t−1)
parity within strict scaling
own bed 1.0–1.6×
11×
115×
0.6–1.2×
Activity for parity
Effective Ru price p(1 − r)/u (USD kg−1)
e
Fe
11.516
fresh compression
+3.988
metal inventory
+1.694
compressor capital
+1.024
refrigeration
+0.000
vessel pressure
−0.787
recycle compression
−0.684
reactor base
−0.117
Ru
16.633
Ru, priced as Fe
10.237
Cost (USD t−1 NH3)
f
bed > 90 m3
economic winner
Fe 653
Ru 347
canonical
Fe feasible
Ru EN draw (eV)
Fe EN draw (eV)
g
Fe feasible, bed ≤ 90 m3
79.2%
Fe economic Top-1
65.3%
Atomic Top-1 = economic Top-1
27.1%
Fe in actionable Top-3
92.3%
Share of 1,000 draws (%)
h
Ru = Fe
median 5.46
Fe cheaper in 5,000 / 5,000 draws
min 1.995
Draws
CRu − CFe (USD t−1 NH3)